

Insect-Pest Management under Natural Farming

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In modern agricultural practices, synthetic insecticides initially protecting the crop and enhancing yields have been countered by pest resistance, loss of beneficial organisms, and environmental contamination. The overreliance on pesticides leads to an array of ecological, economic, and health concerns for farmers due to food residues. The destabilization of natural food chains, rising costs of farming, and alarming health and environmental impacts have triggered a quest for sustainable alternatives. Given these challenges, India is turning to natural farming as a solution, which offers a promising and holistic approach. Shri Subhash Palekar is a key figure in promoting the idea of natural farming, leading to its widespread recognition. It is an ecologically harmonious approach to food production, seeking to minimize human intervention. Rather than relying on synthetic pesticides, fertilizers, or genetically modified organisms, natural farming emphasizes the importance of building healthy soil, maintaining biodiversity, and using traditional and indigenous knowledge systems in combination with modern scientific insights, emphasizing the significance of local resources and the reduction of external inputs.

The evolution of modern agricultural methods, marked by an overwhelming dependence on pesticides and chemical inputs, has witnessed an inadvertent degradation of ecological balance on our planet. Such practices, while initially introduced to boost yield and productivity, have unintentionally resulted in ecological imbalances, as evidenced by disrupted food chains, increased pest resistance and resurgence (Hong-xing et al. 2017). Economically, farmers face the rising expenses, primarily attributed to the high costs of synthetic inputs. Moreover, the health repercussions for both humans and the environment cannot be ignored. The health implications for both the agricultural workers and the larger population are concerning.

In light of these challenges, there is a pressing need for re-examining traditional agricultural practices, and adopting a safer and more sustainable agricultural approach. Therefore, following extensive discussions and opposition, the Indian government has now shifted its focus towards championing "natural farming".

Natural Farming

This farming system is a low-input, climate-resilient approach where the cost of cultivating crops up to harvest is minimal. It emphasizes the use of locally available biodegradable inputs, negating the need for industrial pesticides and synthetic fertilizers, thereby enhancing the farmer's income (Vishnuja and Arunachalam 2020). A Japanese farmer and philosopher, Masanobu Fukuoka (1913–2008), introduced the unique farming concept of natural farming and described the details in his renowned book "The One-Straw Revolution" sometimes referred to as "the Fukuoka method", "natural way of farming," or "do-nothing farming" (Fukuoka 1987). An agriculturist, Shri Subhash Palekar brought the concept of natural farming to India, as Zero Budget Natural Farming (ZBNF), now known as Subhash Palekar Natural farming (SPNF), a contribution for which he was awarded the Padma Shri in 2016. Palekar's approach isn't merely a return to tradition but an intelligent fusion of traditional wisdom with modern scientific knowledge (Chaudhary 2023). At the heart of his method is the emphasis on using locally available resources, minimizing or even eliminating the dependency on expensive external, often costly inputs.

Palekar has outlined procedures to prepare and apply organic mixtures made from locally available resources, known as *astras* to tackle pests and diseases with minimal expenses. He believes that these *astras* not only protect crops from pests and diseases but also neutralize the

damaging residues left behind by conventional pesticides. In addition to *astras*, natural farming integrates sustainable techniques like intercropping for optimized growth, mulching for moisture retention, using resilient local seeds, ensuring minimal soil disturbance, and water management (Palekar 2005). These practices together foster plant health and deter pests, diseases, and weeds. Several Indian states, including Andhra Pradesh, Maharashtra, and Himachal Pradesh, have adopted this method, with many others on the path to embrace it.

Why natural farming?

Natural farming is gaining traction worldwide due to its manifold benefits. A shift toward natural farming is not just about agriculture. It's about re-establishing a sustainable and symbiotic relationship with nature, ensuring food security, and taking proactive steps towards a healthier planet and paving the way for a greener future. The growing realization of the detrimental impacts of synthetic pesticides and fertilizers underscores the pressing need for natural farming. These artificial inputs, while initially enhancing yield, have had a cascading effect on our planet, crops, and human health. Pesticides often persist in the environment, contaminating water sources, diminishing beneficial insect populations, and entering our food chain, where they pose health risks to humans, such as hormonal imbalances, neurotoxic effects, and even carcinogenic outcomes. Synthetic fertilizers, on the other hand, disrupt the natural balance of soil nutrients, leading to reduced soil fertility over time and contributing to phenomena like algal blooms in water bodies due to runoff, which devastates aquatic life. Moreover, the heavy reliance on these inputs has rendered crops more vulnerable to pests and diseases in the long run, weakening their natural resilience. Given these interconnected repercussions, there's an urgent need to embrace natural farming. This method not only mitigates the environment and health risks posed by synthetic chemicals but also advocates for a holistic and sustainable approach to agriculture that prioritizes the well-being of the planet, crops, and humans alike.

Insect-pest management under natural farming

Natural farming maintains a balanced ecosystem, where pest's population is kept at non-damaging levels by natural regulating agents such as predators and parasitoids which provide long-term solution for sustainable agriculture (Devvrat 2018). However, in diverse agro-ecosystems, occasional pest outbreaks can happen due to migration and other factors. The goal here is not to eradicate all pests but to manage them in a way that they don't cause significant damage, while the ecosystem remains in balance. Insect-pest management in natural farming revolves around environment friendly strategies that minimize harm to beneficial organisms and the ecosystem at large. To manage the pest problem, farmers use botanical insecticides derived from local resources. Palekar (2010 & 2011) recommended the dual strategy for plant protection, they first ensure initial defense through seed treatments, followed by regular applications using sprays.

1. Seed Treatment

Seed treatment not only involves applying specific physical, chemical, or biological agents to seeds before sowing to inhibit, manage, or deter pests and pathogens, but it also enhances the germination and vigour of seeds of various crops. Owing to its low-cost technology and minimal environmental impact, seed treatment has become a preferred method for many farmers (DPPQS 2007). In natural farming, a natural product used for seed treatment is known as *beejamrit*.

Beejamrit: In natural farming, "*beejamrit*" emerges as a dual-purpose agent as it acts as a protective seed treatment against insect pests and diseases and enriches the soil with a beneficial microbial community. By treating seeds, seedlings, or other planting materials with *beejamrit* before sowing, one can effectively prevent these seed and soil-borne threats. *Beejamrit* also amplifies the germination potential of seeds, leading to more consistent seedling growth, expedited

root development and enhanced tolerance to varying climatic adversities such as temperature fluctuations, erratic rainfall, and hailstorms (Badiyala and Sharma 2021).

Preparation: Pour 20 liters of water into a plastic tub. Gradually add 5 kg desi (indigenous) cowdung, 5 litres desi cow urine, 250 g burnt lime, and 100 g uncontaminated soil from around a tree base. Stir the mixture gently in a clockwise direction for 2-3 minutes using a wooden stick. Cover the solution with a jute bag and let it remain overnight. The following day, stir the mixture again for 2-3 minutes in a clockwise direction.

Usage: The freshly prepared *beejamrit* should be used within two days and remaining solution after this period should be discarded.

- For seeds: Lay them out on a tarpaulin sheet and sprinkle with *beejamrit* (200 mL per kg of seeds) ensuring they are well-coated. Let them dry in the shade and they'll be ready for sowing next day.
- In cereal crops, place seeds in the palm and apply pressure, ensuring the *beejamrit* coats them well and in pulse crops, gently rub the seeds between fingers to lightly smear them with *beejamrit*. Let the seeds dry overnight and sow them the following morning.
- For vegetatively propagated crops: Place tubers, rhizomes, sets, or grafts in a bamboo basket and immerse the basket in the *beejamrit*-filled tub for 15-20 seconds.
- For transplanted Crops (paddy, vegetables), immerse seedlings in *beejamrit* for 15-20 minutes, then let them dry before transplanting.

2. Natural products application as sprays in pest management

a) **Darekastra/ Paudhastra:** It is an eco-friendly insecticide effective in managing sucking insect-pests and young caterpillars.

Preparation: Chop 2 kg *darek* tree branches with leaves into small pieces. In a barrel, mix 40 litres of water, 2 litres of cow urine, 400 g of cowdung, and the 2 kg of chopped branches. Stir the mixture vigorously for 2-3 minutes clockwise. Allow fermentation by stirring once every few hours for 2 days. After fermentation, strain the solution to remove solids and transfer to a storage barrel. Cover with a jute bag to shield from contaminants and sunlight. The solution can be stored up to 6 months (Badiyala et al. 2023b).

Rate of application: Spray 40 litres in 1 bigha area

b) **Brahmastra:** An organic solution for controlling sucking insect-pests and mature larvae.

Preparation: Mix 200 g crushed leaves of any five plants (*darek*, *papaya*, *guava*, *mango*, *duranta*, *castor*, *custard apple*, *pomegranate*) in a vessel. Add 4 litres of cow urine to the leaves. Cover and simmer on a low flame until it reaches a gentle boil, ensuring the leaf compounds are extracted. Remove from heat and let it cool naturally for 48 hours to intensify potency. Strain out solids and transfer the liquid to a storage container. The *brahmastra* solution can be stored effectively up to 6 months (Sharma et al. 2022).

Rate of application: 1 litre *brahmastra* in 40 litres water for 1 bigha area

c) **Agneyastra:** This organic insecticide targets pests such as root borers, fruit borers and leaf folders which embed themselves within the fruits, roots, and leaves of plants.

Preparation: In a large container, combine 10 litres of desi cow urine, 5 kg of crushed *darek* leaves, 500 g of tobacco powder, 500 g of chilli powder and 500 g of crushed garlic. Gently heat the mixture on a low flame until it begins to boil. Remove it from the heat and allow it to cool for 48 hours. After cooling, strain the mixture through a cotton cloth to remove any solid particles and residues. Store the refined solution in a cool location. When stored properly, the solution remains effective up to six months (Badiyala and Sharma 2021).

Rate of application: 1 litre *agneyastra* in 40 litres water for spraying in 1 bigha area

d) **Dashparni ark:** This solution is designed to effectively manage a wide range of insect pests.

Preparation: Put 4 litres of cow urine, 400 g of cow dung, 100 g of *turmeric* powder, 100 g of *ginger* paste, 5 g of *asafoetida* powder and 200 g of tobacco powder in a barrel and cover the barrel with a jute bag and the following morning, add 200 g of green *chilli* powder, 100 g of *garlic* paste and 400 g of *darek* leaves. Mix the contents using a wooden stick in a clockwise direction for 2-3 minutes. Cover the solution with the jute bag and leave it up to 24 hours.

On the subsequent day, add the following plant leaves, each at 400 g, to the mixture and cover with the jute bag: *lantana*, *dhatura*, *papaya*, *marigold*, *guava*, *bana*, *basuti*, *turmeric*, *ginger*. Stir the mixture in a clockwise direction for 2-3 minutes every morning and evening for a period of 30-40 days. Strain the solution through a cloth and store it. The mixture can be stored and used up to 6 months (Badiyala et al. 2023b).

Rate of application: 1 litre *dashparni ark* in 40 litres of water for spraying in 1 bigha area.

e) **Neemastra:** A natural remedy for sucking pests and young caterpillars on crops

Preparation: Grind 5 kg of *neem* leaves/drupes and place in a drum containing 100 litres of water. Add 5 litres of desi cow urine and 1 kg cowdung. Stir well and cover for 48 hours. Stir thrice over 2 days, then strain through a cloth for spraying (Badiyala and Sharma 2021).

Rate of application: 2 litres in 20 litres of water for 1 bigha area

f) **Neem paste**

It is derived from the *neem* tree (*Azadirachta indica*). Every part of this tree such as seed kernels, flowers, leaves, twigs, or bark has insecticidal qualities, with the seed kernel being the most effective. Azadirachtin, a key component from *neem*, combats several insect species by repelling, inhibiting feeding, deterring oviposition, and disrupting growth. Besides targeting insects, *neem* also fight against nematodes, fungi, bacteria, and mollusks.

Preparation: Combine 30 litres of water, 20 litres of cow urine, 20 kg of cow dung, 20 kg of chopped *neem* leaves in a drum. Mix thoroughly and cover for 48 hours, stirring occasionally. After 2 days, the paste is ready and should be used within a week of making.

Time of application: Apply on the trunk of fruit trees twice a year *i.e.* end of May and October.

g) **Fermented butter milk**

Preparation: Keep the butter milk prepared from milk of desi cow in sunlight in a container and let it ferment. This process may take 4-6 days in summers and up to 10 days in winters. As soon as the froth starts appearing on butter milk, it is ready to use. In a drum (100 litres capacity), add 3 litres fermented butter milk and water so that the total solution becomes 100 litres. Stir the solution with a wooden stick in clockwise direction for 2-3 minutes and use it right away. While it is mainly used as a fungicide, it can also help manage insect pests.

h) **Lantana extract**

Locally called “Panchphuli”, *lantana*’s leaf extract is efficacious against beetle pests, potato tuber moth, borers and sucking pests attacking different crops.

Farming practices for effective pest management

- **Regular Monitoring:** Regular inspection of plants and the surrounding environment helps in early detection of any potential pest problems, allowing for timely interventions.
- **Crop Rotation:** By regularly changing the crops grown in a particular field, the life cycle of pests can be disrupted, reducing their populations over time.
- **Intercropping & Polyculture:** Growing different crops together can confuse pests and make it difficult for them to spread rapidly. Certain plants can also act as repellents for specific pests (Devvrat 2018).

- Mulching: Organic mulches can prevent certain pests, from reaching plants. They can also create habitats for beneficial insects.
- Cultural Practices: Simple practices like removing affected plant parts or maintaining proper spacing can reduce the spread of pests.
- Natural biological control by conserving the potential biological agents can be an important tool in pest management under natural farming.

Important research findings on the management of insect-pests and mite under natural farming

Natural products and biopesticide formulations serve as biological insecticides, each with distinct modes of action. They can be effectively incorporated into pest management strategies either individually or in combination. The adverse impacts of plant products on insects can manifest in various ways, such as inducing toxicity, causing mortality, acting as antifeedants, inhibiting growth, suppressing reproductive behaviour, and reducing fecundity and fertility (Sharaby and Nojiban 2015). The following are some studies related to the application of these natural products in pest control:

Table 1. Use of natural products in pest management

Bio-formulations	Conc. Used	Effective against	Host crop	References
<i>Agneyastra</i>	5%	<i>Spodoptera litura</i> ,	Tomato	Badiyala and Kanwar (2021)
		<i>Chromatomyia horticola</i> ,	Garden pea	Sharma et al. (2014b)
		<i>Liphaphis erysimi</i>	Okra	Sharma et al. (2014a)
<i>Neemastra</i>	10%	Sucking pests	<i>Bt cotton</i>	Patel et al. (2017b)
<i>Dashparni ark</i>	5%	<i>Myzus persicae</i>	Capsicum	Sharma et al. (2014a)
<i>Brahmastra</i>	20%	Sucking pests	<i>Bt cotton</i>	Patel et al. (2017b)
<i>Darekastra</i>	10%	Red spider mite (<i>Tetranychus urticae</i>)	Cucumber	Thakur and Sood (2022)
	5%	<i>Spodoptera litura</i>	Tomato	Badiyala and Kanwar (2021)
<i>Tamarlassi</i>	10%	Greenhouse whiteflies	Cucumber	(Kumari and Sood 2023).
<i>Agneyastra – Darekastra</i> (alternate sprays)	10%	Okra leafhopper, (<i>Amrasca biguttula biguttula</i>)	Okra	Badiyala et al. (2023a)

- Research by Carpinella et al. (2003) demonstrated that limonoids, found in the fruit extract of *Melia azedarach*, disrupt certain physiological processes in insects, acting as a deterrent and toxin. This extract was particularly effective against pests from families such as Curculionidae, Tenebrionidae, and Chrysomelidae.

- Using a *dashparni* extract (5%), the mortality rate of *Myzus persicae* in capsicum progressively increased over time, recording 5.44 per cent at 24 hours, 13.52 per cent at 48 hours and 16.52 per cent at 72 hours (Sharma et al. 2014a).
- A significant reduction in *Leucinodes orbonalis* was observed by Mishra et al. (2014) after treatment with a combination of vermiwash and a biopesticide. This biopesticide comprised *neem* (*Azadirachta indica*) oil, aqueous extract of *garlic* (*Allium sativum*), and leaf extract of custard apple (*Annona squamosa*).
- Three consecutive applications of *agneyastra* (5%) or *neem* oil (0.03%), at 10-days intervals were found to be effective in managing *Chromatomyia horticola* (pea leaf miner) infestations in peas (Sharma et al. 2014b).
- Condensed cow urine is known to exhibit insecticidal, antibacterial and antifungal properties (Prashith et al. 2014). It can be used on its own or combined with other biopesticides (Mudigoudra et al. 2009). Patel et al. (2017a) found that both cow urine (20%, 50%) and *neem* leaf extract (10%) were effective in combating aphids. Furthermore, their research indicated that plots treated with cow urine (50%) exhibited the lowest (1.40 aphids per leaf) population of *Aphis gossypii*, followed by *vermiwash* (50%) with 1.81 aphids per leaf.
- Azadirachtin (3mL/L) and *darekastra* (100mL/L) was promising in suppressing the population of sucking pests like aphids (*Aphis gossypii*) and jassids (*Amrasca biguttula biguttula*) infesting okra in Himachal Pradesh (Kaushal et al. 2021).
- Fermented buttermilk and *tamarlassi* has shown lethal and inhibitory activities the red spider mite adults and *darekastra*, *jeevamrit*, and cow urine were also moderately effective against the pest (Thakur and Sood 2019).
- Yankit et al. (2019) examined the incidence of *Phthorimaea absoluta* across three farming methods: zero budget natural farming (ZBNF), organic farming, and conventional farming. Their findings showed that ZBNF had the lowest pest incidence, with the pest appearing 4 weeks later than in conventional plots. In ZBNF, alternate weekly applications of *darekastra* and *brahmastra* provided optimal control.
- A study on the feeding inhibition effects of bioformulations like *neemastra*, *agneyastra*, *brahmastra*, *dashparni ark*, and a mix of *garlic + ginger + mint* against *Agrotis ipsilon* revealed *dashparni ark* as the most potent. It notably hindered the feeding of later instar larvae, outperforming the other formulations. Specifically, *dashparni ark* was 1.35 times more effective than the mixture of *garlic*, *ginger*, and *mint* (Joshi et al. 2020).
- A regimen of weekly sprays combining *darekastra* (5%) and *agneyastra* (5%) was successful in reducing the larval population of *Spodoptera litura* on tomato plants under polyhouse conditions (Badiyala and Kanwar 2021).
- *Agneyastra* (800 mL/10L), followed by *neemastra* (400L/acre) and *brahmastra* (800 mL/10L), significantly reduced root knot nematode infestation when drenched around tomato plant roots during transplanting and re-applied at 15, 30, and 45-day intervals (Maru et al. 2021).
- In the third week of March, the peak population of aphids, specifically *Rhopalosiphum padi* and *Sitobion avenae*, was observed to be highest in conventional farming (29.65 aphids per shoot) followed by organic farming (24.18 aphids per shoot), while natural farming registered the lowest at 21.40 aphids per shoot. Additionally, the population of coccinellid predators was highest in conventional farming with an average of 7.10 adults per 10 plants /m², slightly more than the 7 adults per 10 plants/ m² found in natural farming (Thakur 2021).
- Cow-derived products like *darekastra* and *panchagavya* are known to have larvicidal properties against *S. litura* larvae. *Darekastra* caused 76.67 per cent larval mortality and significantly reduced populations by 59 per cent within three days. These products could be alternative solutions to traditional pest management. (Anees and Chandel 2022).

- Seasonal abundance of cauliflower pests viz., aphids, flea beetles and whiteflies on two farming systems resulted in consistently lower pest counts in zero budget natural farming (ZBNF) as compared to the conventional farming (CF). However, for the natural enemies such as *Diaeretiella rapae*, ZBNF had higher parasitizations, compared to conventional farming. Similarly, coccinellid beetles and syrphids were more prevalent in ZBNF, with *Diadegma semiclausum* and *Cotesia glomerata* exclusively observed in ZBNF, peaking notably in the 5th and 9th weeks, respectively (Barakzai et al. 2022).
- *Darekastra* and *tamarlassi* used at a 10 per cent concentration, showed a moderate capability to combat red spider mite, *Tetranychus urticae* in cucumber polyhouses. When applied as a preventative measure, they decreased mite infestation and improved economic gains. Notably, *darekastra* yielded the most substantial returns (Thakur and Sood 2022).
- Various natural products were evaluated against *Amrasca biguttula biguttula* infesting okra under natural farming conditions. Among the various treatments, four weekly sprays of *agneyastra* 10 per cent and *darekastra* 10 per cent sprays when used in rotation proved best in effectively checking the population of okra leafhoppers followed by *agneyastra* 10 per cent and *neemastra* 10 per cent sprays (Badiyala et al. 2023a)
- Natural products were tested for their efficacy against greenhouse whiteflies on cucumber plants under polyhouse conditions. In preventive treatments, *tamarlassi* (10%) was the most effective, with a population of 0.06 adults per three leaves followed by fermented cow urine (5%) and fermented buttermilk (10%). For curative measures, *tamarlassi* (10%) remained the most effective, reducing the whitefly population by 67.99 per cent followed by fermented cow urine (5%) and a combination of *vermiwash* and *tamarlassi* (10%), with reduction of 63.94 and 63.35 per cent, respectively (Kumari and Sood 2023).
- Various treatments were applied against brinjal shoot and fruit borer, *L. orbonalis*, 20 days post-transplanting until harvest. Cow urine and *agneyastra* had the least fruit infestation at 4.43 per cent and 5.56 per cent, respectively. By weight, *agneyastra* (4.13%), cow urine (4.21%), and *brahmastra* (5.20%) had minimal infestations. *Brahmastra* and *agneyastra* yielded the highest fruit yield of 45.50 q/ha and 41.67 q/ha, respectively (Sood et al. 2023).

Conclusion

Natural farming offers an avenue towards a more sustainable and resilient agricultural future. One of its most notable advantages lies in its approach to pest management. By harnessing the equilibrium of nature and utilizing beneficial organisms and plants, natural farming minimizes the need for chemical pesticides, which are detrimental to both the environment and human health. This method not only protects crops but also helps preserve biodiversity, maintaining the delicate balance of ecosystems. Enhancing soil fertility, promoting water retention, and reducing erosion further underscore its merits. However, it's essential that thorough research and experiments are conducted to understand the true potential of natural pest management. By educating farmers about these benefits and demonstrating the ease of adopting natural solutions, we can inspire a broader shift towards these practices. This evolution in agriculture not only positions India at the forefront of sustainable farming but also signals a global trend toward a healthier, more balanced relationship with our environment.

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